

DS701 Homework 4: Probability, MLE, and EM

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SOLUTIONS

Problem 1: A screening test, three ways

Practices: Bayes' rule with the base rate, and the odds / likelihood-ratio form.

A screening test for a disease D has **sensitivity** $P(+ | D) = 0.90$ (positive in 90% of people who have D) and **specificity** $P(- | \bar{D}) = 0.90$ (negative in 90% of people who do not). Bayes' rule for events, with the denominator expanded by the law of total probability (with $C_k \in \{D, \bar{D}\}$):

$$P(D | +) = \frac{P(+ | D) P(D)}{P(+ | D) P(D) + P(+ | \bar{D}) P(\bar{D})}.$$

- (a) In the general population the **prevalence** (base rate) is $P(D) = 0.01$. A randomly chosen person tests positive. Compute $P(D | +)$, showing the two terms of the denominator separately. In one sentence, why is the answer so far below the 90% that “the test is 90% accurate” suggests? $P(D | +) = \underline{\hspace{2cm}}$
- (b) The same test is used in a high-risk clinic where $P(D) = 0.10$. Recompute $P(D | +)$. Then state the practical lesson in one or two sentences: the test did not change — what did, and what does that say about using one test both for mass screening and in a specialist clinic? $P(D | +) = \underline{\hspace{2cm}}$
- (c) **Odds form.** Define the odds of an event as $O(D) = P(D)/P(\bar{D})$ and the positive likelihood ratio as $LR_+ = P(+ | D)/P(+ | \bar{D})$. Show, starting from Bayes' rule written once for D and once for $\bar{D}$, that

$$O(D | +) = LR_+ \times O(D)$$

(posterior odds = likelihood ratio $\times$ prior odds — no denominator needed). Optionally, then compute LR_+ for this test, redo (a) and (b) in odds and convert back to probabilities to confirm your answers, and finally: a person from the general population tests positive on **two independent** administrations of the test — what is $P(D | ++)$? (Independence here means the second test multiplies the odds by LR_+ again.) $LR_+ = \underline{\hspace{2cm}}$ $P(D | ++)$ = $\underline{\hspace{2cm}}$

Solution

- (a) $P(+ | D)P(D) = 0.9 \times 0.01 = 0.009$; $P(+ | \bar{D})P(\bar{D}) = 0.1 \times 0.99 = 0.099$; $P(+)$ = 0.108. So $P(D | +) = 0.009/0.108 = 1/12 \approx 0.083$. The false positives (0.099) outnumber the true positives (0.009) eleven to one because the disease is rare: 10% of a large healthy

population is far more people than 90% of a tiny sick one. “90% accurate” describes $P(+ | D)$, not $P(D | +)$ — the base rate is what separates them.

- (b) $0.9 \times 0.1 = 0.09$; $0.1 \times 0.9 = 0.09$; $P(D | +) = 0.09/0.18 = 0.5$. Only the prior changed, from 0.01 to 0.10, and the posterior went from 0.083 to 0.5. A positive result means very different things in the two settings: in mass screening it is mostly a false alarm and should trigger a confirmatory test, not a diagnosis; in the clinic, where the prior is high, the same result is a coin flip and worth acting on. (Common mistake: reporting 0.9 in both cases — that ignores the base rate entirely.)
- (c) $P(D | +) = P(+ | D)P(D)/P(+)$ and $P(\bar{D} | +) = P(+ | \bar{D})P(\bar{D})/P(+)$. Divide the first by the second; $P(+)$ cancels: $O(D | +) = \frac{P(+ | D)}{P(+ | \bar{D})} \cdot \frac{P(D)}{P(\bar{D})} = LR_+ \times O(D)$. Here $LR_+ = 0.9/0.1 = 9$. (a): prior odds $0.01/0.99 = 1:99$; posterior odds $9:99 = 1:11$; $P = 1/(1 + 11) = 1/12 \approx 0.083$. (b): prior odds $1:9$; posterior odds $9:9 = 1:1$; $P = 0.5$. Two independent positives: odds $1:99 \times 9 \times 9 = 81:99$; $P(D | ++) = 81/180 = 0.45$. Each positive multiplies the odds by 9, but a 1:99 start needs two of them just to reach even odds — the same base-rate effect, now in a form where repeated evidence is a simple product.

Problem 2: Maximum likelihood, from scratch

Practices: the MLE recipe — likelihood, log, derivative, solve — on two models, and knowing why the log is there.

- (a) **Bernoulli.** Each observation is $x_n \in \{0, 1\}$ with $P(x_n = 1) = p$, so $P(x_n | p) = p^{x_n}(1-p)^{1-x_n}$. For N i.i.d. observations with $k = \sum_n x_n$ ones: write the likelihood $L(p)$, the log-likelihood $\ell(p)$, set $d\ell/dp = 0$, and solve for $\hat{p}$. Then evaluate $\hat{p}$ for the data 1, 0, 1, 1, 0, 1, 1, 1, 0, 1. (Say in one line why the critical point is a maximum and not a minimum.) For the data, $\hat{p} =$

- (b) **Gaussian mean and variance.** Start from the Gaussian log-likelihood

$$\ell(\mu, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2.$$

Derive $\hat{\mu}$ from $\partial\ell/\partial\mu = 0$. Then treat $v = \sigma^2$ as the variable, derive $\hat{\sigma}^2$ from $\partial\ell/\partial v = 0$ (substituting $\hat{\mu}$), and evaluate both on the data 2, 3, 5, 6, 9. Compare $\hat{\sigma}^2$ with the sample variance s^2 that divides by $N - 1$ — which is larger, and by what factor here? $\hat{\mu} =$ _____
 $\hat{\sigma}^2 =$ _____ $s^2 =$ _____ factor: _____

- (c) **Why the log?** Give the two reasons the lecture gives for maximizing ℓ rather than L : one about *where* the maximum is, one about the *form* of the function. Then add the practical one: for $N = 2000$ observations from (a), every factor of L is at most 1 — say roughly how small L is at best (e.g. compare 0.5^{2000} with the smallest positive double, about 10^{-308}) and why ℓ does not have this problem.

Solution

- (a) $L(p) = \prod_n p^{x_n} (1-p)^{1-x_n} = p^k (1-p)^{N-k}$; $\ell(p) = k \log p + (N-k) \log(1-p)$; $\frac{d\ell}{dp} = \frac{k}{p} - \frac{N-k}{1-p} = 0 \Rightarrow k(1-p) = (N-k)p \Rightarrow \hat{p} = k/N$. It is a maximum because $\ell'' = -k/p^2 - (N-k)/(1-p)^2 < 0$ (or: $\ell \rightarrow -\infty$ at both ends $p \rightarrow 0, 1$, so the single critical point is the peak). Data: $k = 7$, $N = 10$, $\hat{p} = 0.7$.
- (b) $\partial \ell / \partial \mu = \frac{1}{\sigma^2} \sum_n (x_n - \mu) = 0 \Rightarrow \hat{\mu} = \frac{1}{N} \sum_n x_n$ (matches the lecture's one-line derivation). With $v = \sigma^2$: $\ell = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log v - \frac{1}{2v} S$, where $S = \sum_n (x_n - \hat{\mu})^2$; $\partial \ell / \partial v = -\frac{N}{2v} + \frac{S}{2v^2} = 0 \Rightarrow \hat{v} = S/N$, i.e. $\hat{\sigma}^2 = \frac{1}{N} \sum_n (x_n - \hat{\mu})^2$. Data: $\hat{\mu} = 25/5 = 5$; deviations $-3, -2, 0, 1, 4$, squares $9, 4, 0, 1, 16$, $S = 30$; $\hat{\sigma}^2 = 30/5 = 6$; $s^2 = 30/4 = 7.5$. The $N-1$ version is larger by the factor $N/(N-1) = 5/4$.
- (c) (i) log is monotonically increasing, so ℓ and L have their maximum at the *same* θ — nothing is lost. (ii) The log turns the product over observations into a sum, so the derivative is a sum of simple terms and the algebra (as in (a) and (b)) is short. (iii) Numerically: L is a product of thousands of numbers ≤ 1 ; even in the best case $0.5^{2000} = 10^{-602}$, far below 10^{-308} , so in floating point L underflows to exactly 0 for every p and cannot be maximized. $\ell = 2000 \log 0.5 \approx -1386$ is an ordinary number; sums of logs stay in range.

Problem 3: One EM iteration by hand

Practices: the E-step and M-step formulas on numbers small enough to see what they do, and the argument for why EM behaves.

A one-dimensional mixture of $K = 2$ Gaussians, both with variance $\sigma^2 = 1$ (**keep the variances fixed at 1 throughout** — update only means and weights). Current parameters: $\mu_1 = 0$, $\mu_2 = 2$, $\pi_1 = \pi_2 = 0.5$. Data: $x_1 = 0$, $x_2 = 1$, $x_3 = 2$, $x_4 = 4$. Useful values: $e^{-0.5} \approx 0.607$, $e^{-2} \approx 0.135$, $e^{-8} \approx 0.00034$; note that with equal variances the factor $1/\sqrt{2\pi}$ is the same in every term and cancels in the responsibilities.

- (a) **E-step.** For each data point compute the responsibilities γ_{i1}, γ_{i2} (3 decimals). Before computing, say for which point the answer must be exactly 0.5 and why. (γ_{i1}, γ_{i2}) for $x_1, \dots, x_4$: _____
- (b) **M-step.** Compute N_1, N_2 , the new means μ'_1, μ'_2 and the new weights π'_1, π'_2 . In one sentence each: why did μ_2 move more than μ_1 , and why did the weights become unequal? $N_1 =$ _____ $N_2 =$ _____ $\mu'_1 =$ _____ $\mu'_2 =$ _____ $\pi'_1 =$ _____ $\pi'_2 =$ _____
- (c) The lecture states that the log-likelihood $\ell(\theta)$ never decreases from one EM iteration to the next. Explain the mechanism in words: what the E-step computes, what the M-step maximizes and why that has a closed form, and why alternating the two cannot lower ℓ . Then say what this guarantees about the sequence $\ell^{(t)}$, what it does **not** guarantee, and what is done about that in practice (relate to Homework 3, Problem 1).
- (d) **Hard vs. soft.** Without recomputing, describe what the four responsibilities in (a) would look like if both variances were $\sigma^2 = 0.01$ instead of 1, and what the M-step for the means

would then reduce to. Which algorithm is that? State the one practical difference and the one conceptual difference between GMM/EM and k -means that the lecture draws.

Solution

- (a) With equal π_k and equal variances, $\gamma_{i1} = \frac{e^{-(x_i-0)^2/2}}{e^{-(x_i-0)^2/2} + e^{-(x_i-2)^2/2}}$. The point $x_2 = 1$ is equidistant from the two means, so both numerators are $e^{-0.5}$ and $\gamma_{21} = \gamma_{22} = 0.5$ exactly. $x_1 = 0$: $e^0 = 1$ vs. $e^{-2} = 0.135$: $\gamma_{11} = 1/1.135 = 0.881$, $\gamma_{12} = 0.119$. $x_2 = 1$: 0.5, 0.5. $x_3 = 2$: e^{-2} vs. e^0 : $\gamma_{31} = 0.119$, $\gamma_{32} = 0.881$. $x_4 = 4$: $e^{-8} = 0.00034$ vs. $e^{-2} = 0.135$: $\gamma_{41} = 0.00034/0.13534 = 0.0025$ (call it 0.002–0.003), $\gamma_{42} = 0.9975$. Each row sums to 1. (Common mistake: forgetting the square in the exponent, or using $e^{-(x-\mu)^2}$ instead of $e^{-(x-\mu)^2/2}$.)
- (b) $N_1 = 0.881 + 0.5 + 0.119 + 0.0025 = 1.5025 \approx 1.503$; $N_2 = 0.119 + 0.5 + 0.881 + 0.9975 = 2.4975 \approx 2.498$ (they sum to $n = 4$). $\mu'_1 = (0.881 \cdot 0 + 0.5 \cdot 1 + 0.119 \cdot 2 + 0.0025 \cdot 4)/1.5025 = 0.748/1.5025 \approx 0.498$. $\mu'_2 = (0.119 \cdot 0 + 0.5 \cdot 1 + 0.881 \cdot 2 + 0.9975 \cdot 4)/2.4975 = 6.252/2.4975 \approx 2.503$. $\pi'_1 = 1.5025/4 \approx 0.376$, $\pi'_2 = 2.4975/4 \approx 0.624$. μ_2 moved more because the far point $x_4 = 4$ is almost entirely component 2's responsibility and drags its weighted mean right, while component 1 only picks up the half-share of $x_2 = 1$. The weights became unequal because component 2 now “owns” more total responsibility ($N_2 > N_1$) — three of the four points lie at or to the right of the midpoint 1. (Check: the log-likelihood rises from -7.999 to -7.315 .)
- (c) The E-step computes, for the current θ , the posterior probability $\gamma_{ik} = P(z_i = k | x_i, \theta)$ that each point came from each component — Bayes' rule with π_k as prior and $\mathcal{N}(x_i | \mu_k, \Sigma_k)$ as likelihood. Given those soft assignments, the M-step maximizes the *expected complete-data* log-likelihood $\sum_i \sum_k \gamma_{ik} [\log \pi_k + \log \mathcal{N}(x_i | \mu_k, \Sigma_k)]$; because the log now sits on individual Gaussians (not on a sum of them), the derivatives separate and the maximizers are the weighted MLE formulas — the closed form. Why ℓ cannot go down: with the responsibilities fixed at their E-step values, that expected complete-data objective (plus a term that does not depend on θ) is a lower bound on $\ell(\theta)$ that *touches* ℓ at the current θ ; the M-step raises the bound, and ℓ sits above the raised bound, so $\ell(\theta^{\text{new}}) \geq \ell(\theta^{\text{old}})$ (the lecture asserts the monotone increase; the “touching lower bound” sentence is the reason). Guarantee: $\ell^{(t)}$ is non-decreasing and bounded above, so it converges — which is what the log-likelihood-change stopping rule relies on. Not guaranteed: that the limit is the global maximum; EM finds a *local* maximum and depends on the initialization — exactly like the k -means run in HW3 Problem 1(c) that stopped at WCSS 74 instead of 33.5. In practice: initialize from k -means and/or run several restarts, keep the highest ℓ .
- (d) With $\sigma^2 = 0.01$ the exponents are 100 times larger, so every responsibility saturates: x_1 is $\approx (1, 0)$, x_3 and x_4 are $\approx (0, 1)$, and only the exactly-equidistant $x_2 = 1$ stays at $(0.5, 0.5)$ (a knife-edge tie). With 0/1 responsibilities, $\mu'_k = \frac{1}{N_k} \sum_i \gamma_{ik} x_i$ is just the plain mean of the points assigned to component k — the k -means update; the E-step is the nearest-center assignment. So EM with tiny, equal, fixed variances *is* Lloyd's algorithm. Practical difference (lecture): GMM makes soft assignments (“soft k -means”) where k -means makes hard ones. Conceptual difference: the GMM states an explicit generative assumption — data drawn from a mixture of Gaussians with their own covariances — which can be examined and replaced; k -means only declared an objective, and its implicit spherical/equal-variance assumption was never stated.